

Separability-Aware Synthetic Data Selection for Rare-Outcome Prediction in Health Data

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Abstract

Background: Synthetic data are used to increase positive-class sample size in rare-outcome prediction, but additional records may not improve prediction when classes overlap or generation fails to preserve class-conditional structure.

Objective: We tested whether post-generation control of class separability improves held-out-real utility beyond class-frequency correction and whether any gain is accompanied by reduced positive-class diversity.

Methods: SEPAWARE ranks an oversized synthetic candidate pool using nearest-real proximity, nearest-hit/nearest-miss margin, and probability from a real-trained classifier while preserving prespecified class counts. We evaluated Vigitel obesity, COVID-19 hospital mortality, Vigitel current smoking, and kidney-cohort 365-day mortality; the two Vigitel tasks share one source. Three repeated 5-fold partitions compared real-only training, random over- and under-sampling, SMOTE, class weighting, CTGAN, TVAE, and their selected variants. AUPRC was the primary augmentation endpoint. TSTR, calibration, fidelity, positive-class coverage and diversity, exact matches, and a membership proxy were reported separately. Primary paired tests used a corrected repeated-cross-validation procedure and Holm adjustment.

Results: Mean AUPRC differences between SEPAWARE and standard CTGAN were 0.012 for COVID-19, 0.009 for Vigitel obesity, 0.031 for Vigitel smoking, and 0.005 for kidney mortality. Only smoking excluded zero after repeated-cross-validation correction and Holm adjustment (95% CI 0.015–0.048; adjusted $p = .009$). All four TVAE intervals included zero. Selected CTGAN did not exceed the strongest conventional comparator on any task. CTGAN selection reduced positive-class coverage distance in every task but also reduced diversity. Exact-match rates were at most 0.0005 and membership-proxy AUROCs were near 0.5; these are not privacy guarantees.

Conclusions: Class-boundary control differs from class balancing, but the evidence supports a generator-relative augmentation gain only for the smoking task. Selection should be compared with simple baselines and assessed for calibration and loss of positive-class heterogeneity. Internal cross-validation does not establish clinical benefit, transportability, or privacy.

Keywords: synthetic health data; tabular data; class imbalance; class overlap; rare outcomes; data augmentation; privacy; calibration

1 Introduction

Health datasets combine sensitive information, mixed numerical and categorical variables, missingness, and outcomes that are often uncommon but clinically consequential. Synthetic tabular data can support model development and collaboration when access to real records is restricted, but generation alone does not guarantee privacy or downstream usefulness [1–3]. In particular, a generator may preserve aggregate distributions while weakening the class-conditional structure needed to identify a rare outcome or may yield deceptively easy synthetic examples [4, 5].

Rare-outcome learning is commonly framed as a class-imbalance problem. Random resampling, the Synthetic Minority Over-sampling Technique (SMOTE), class weighting, and conditional generation increase the number or influence of positive-class observations [6, 7]. More data, however, need not provide more information when positive and negative observations overlap in feature space. Class prevalence and class overlap are distinct properties: data-level or algorithm-level imbalance correction changes representation or decision cost, whereas it does not necessarily repair weak class-conditional geometry [7].

Existing synthetic-health-data evaluations typically assess global resemblance, downstream utility, and privacy after generation [1, 2, 5]. They provide less guidance on whether post-generation control of class geometry improves rare-outcome prediction, when such control fails, and whether apparent gains arise by discarding difficult but legitimate positive cases. This gap motivates the present study. Our objective is to determine whether separability-aware candidate selection adds predictive value beyond class-count correction and standard generation, while explicitly measuring calibration, positive-class coverage and diversity, and empirical privacy risk.

We developed SEPAWARE, a post-generation mechanism that draws an oversized candidate pool and ranks records using proximity to the real training manifold and label-consistent geometric evidence. It retains a prespecified class count, separating the effect of candidate selection from a change in class ratio. We first isolate the mechanism in Vigitel obesity and COVID-19 hospital mortality, then evaluate practical augmentation across those tasks plus Vigitel current smoking and landmark-defined kidney-cohort mortality.

In the completed augmentation benchmark, mean AUPRC differences between SEPAWARE and standard CTGAN were 0.012 for COVID-19 mortality, 0.009 for Vigitel obesity, 0.031 for Vigitel smoking, and 0.005 for kidney mortality. After correction for repeated cross-validation and Holm adjustment, only the smoking contrast excluded zero. TVAE differences were small and inconsistent, and simple resampling or real-only training remained competitive. These results test the selection rule without assuming that it will improve every task.

1.1 Research Questions

The study addresses three research questions:

- RQ1:** Is data-level or algorithm-level class-imbalance correction sufficient, or does class overlap limit the utility of synthetic augmentation?
- RQ2:** Does separability-aware selection improve rare-outcome prediction beyond random sampling, SMOTE, class weighting, and unconstrained generation?
- RQ3:** When predictive performance improves, is the gain accompanied by acceptable calibration, positive-class coverage and diversity, and empirical privacy risk?

The mechanism experiment uses leakage-controlled synthetic-to-real evaluation followed by a blocked 2×2 factorial analysis that separates separability weighting from clinical-anchor weighting. The common benchmark evaluates real-plus-synthetic augmentation and TSTR on shared held-out real folds using conventional imbalance methods, CTGAN, TVAE, and SEPAWARE variants. AUPRC is the primary augmentation endpoint; secondary analyses include Macro-F1, positive-class metrics, AUROC, balanced accuracy, Brier score, calibration, coverage, diversity, fidelity, and privacy-risk proxies.

1.2 Study scope

The work has four parts:

1. It frames class overlap as a target distinct from class imbalance and global fidelity in synthetic health-data augmentation.
2. It introduces an auditable post-generation selection mechanism whose effect can be separated from generator fitting and class-count correction.
3. It uses a factorial experiment to attribute the strong Vigitel obesity synthetic-to-real change to separability weighting while reporting the COVID-19 null attribution with equal emphasis.
4. It tests whether selection concentrates easy cases through a common benchmark spanning four outcome tasks, two generator families, conventional imbalance baselines, calibration, positive-class structure, and privacy diagnostics.

The remainder of the paper reviews synthetic tabular health data and rare-outcome evaluation, defines the datasets and leakage-controlled protocols, reports common and dataset-specific results, and concludes with limitations and requirements for external validation.

2 Related Work

This section positions SEPAWARE within four connected areas: synthesis of mixed-type health tables, evaluation beyond aggregate resemblance, rare-outcome learning under imbalance and overlap, and the privacy–utility tension. Each area supplies part of the required evaluation, but their intersection remains underexplored.

2.1 Synthetic Tabular Health Data

Tabular health data represent one person or encounter per row and heterogeneous clinical or survey variables in columns; synthetic microdata are artificial row-level records intended to reproduce selected properties of such data without directly reproducing the source rows. Statistical microdata synthesis predates deep learning. Multiple imputation and probabilistic graphical models generate records from estimated conditional distributions and remain attractive because their assumptions can be inspected [8]. Copulas provide efficient dependence modelling but can struggle when discrete, continuous, multimodal, and rare-event variables interact nonlinearly [9].

Generative adversarial networks formulate generation as an adversarial game [10]. MedGAN adapted this approach to high-dimensional discrete patient records [11], and CTGAN addressed mixed-type tables through mode-specific normalization, conditional vectors, and training-by-sampling [12]. Variational autoencoders optimize a likelihood-related lower bound [13]; TVAE is a tabular extension evaluated alongside CTGAN [12]. TabDDPM later applied separate diffusion processes to numerical and categorical attributes and outperformed several GAN and VAE comparators on general tabular benchmarks [14]. These methods improve generation, whereas SEPAWARE operates after generation: it tests whether transparent control of the candidate pool’s class geometry can improve rare-outcome utility. We use CTGAN and TVAE to examine transfer across two generator families; diffusion remains a future comparator.

2.2 Evaluation Beyond Aggregate Resemblance

Health-synthesis studies repeatedly conclude that no single metric establishes usefulness or safety and that evaluation should jointly cover resemblance, downstream utility, and privacy [1, 2, 5]. Recent comparisons report dataset- and metric-dependent rankings and a persistent fidelity/utility–privacy trade-off [15, 16]. A review of structured synthetic electronic health records also documents continuing problems with rare events, reproducibility, and privacy evaluation [4].

Univariate fidelity measures whether synthetic columns resemble their real counterparts, pairwise fidelity evaluates dependence structure, and multivariate distinguishability tests whether a classifier separates real from synthetic records [1, 2]. None alone establishes rare-outcome usefulness: aggregate scores can be dominated by negative-class records, whereas a selector can improve classification by producing easy positive prototypes rather than a faithful positive population [4, 5].

Train-on-Synthetic, Test-on-Real (TSTR) evaluation probes whether relations learned from synthetic records transfer to held-out real observations; Real-plus-Synthetic-to-Real evaluation instead tests augmentation [1, 2]. These regimes answer different questions and are reported separately here. Neither constitutes a privacy guarantee or proves unbiased clinical inference.

2.3 Class Imbalance, Overlap, and Augmentation

Accuracy can remain high while positive-class recall is poor in imbalanced data [7]. Macro-F1 gives both classes equal weight, while AUPRC is often more informative than AUROC for rare positive outcomes [17]. Conventional approaches include random resampling, SMOTE [6], cost-sensitive learning, and threshold adjustment [7]. These methods alter frequency or decision cost but do not necessarily recover informative class-conditional structure.

A recent comparison across 13 small health datasets found that sampling with replacement could match more complex generative augmentation in several settings, suggesting that some apparent gains arise mainly from increased sample size [18]. This makes random oversampling a necessary comparator and sharpens our question: does controlling generated class geometry add value beyond correcting sample counts?

Class imbalance and class overlap are related but distinct [7]. A rare positive class can remain geometrically well separated, whereas a less imbalanced outcome can be intrinsically difficult because valid observations occupy overlapping feature regions. Adding examples cannot remove irreducible overlap, and aggressively selecting distant examples may improve an empirical boundary by deleting difficult but legitimate cases. We therefore interpret discrimination jointly with calibration and positive-class coverage and diversity.

2.4 Privacy–Utility Tension

Synthetic data are not anonymous by construction: high-capacity generators can reproduce or reveal information about training records [3, 19]. Privacy can be examined through nearest-record, membership-inference, and attribute-inference analyses or protected through formal mechanisms such as differential privacy [20]. LDP-GAN combines local differential privacy with medical-record synthesis [21], while a recent oncology framework evaluates candidate synthetic datasets across multiple quality dimensions before downstream use [22]. Accordingly, our study reports empirical privacy proxies separately from fidelity and utility and does not equate a near-chance proxy with anonymity.

2.5 Research Gap and Study Positioning

Table 1 summarizes how representative studies address the dimensions central to this work. The literature leaves a practical gap at the intersection of rare outcomes, explicit class-overlap control, held-out-real augmentation, positive-class coverage, and privacy diagnostics. Generators optimize distribution learning, imbalance methods correct representation or decision cost, and evaluation frameworks generally assess outputs after generation rather than steering class-conditional geometry. Our contribution is not a universally superior generator; it is an auditable intervention and evaluation design for determining when separability-aware selection helps and what it may sacrifice.

Table 1: Positioning of representative related work. A checkmark denotes an explicit focus; a dash denotes that the dimension is not the study’s primary contribution.

Study	Health	Rare outcome	Overlap	Utility	Privacy
MedGAN [11]	✓	–	–	✓	–
CTGAN/TVAE [12]	–	–	–	✓	–
Health-data evaluation [1]	✓	–	–	✓	✓
Structured-EHR review [4]	✓	✓	–	✓	✓
LDP-GAN [21]	✓	–	–	✓	✓
This study	✓	✓	✓	✓	✓

3 Materials and Methods

This section defines the four outcome tasks, leakage-controlled evaluation regimes, generators, selection rule, factorial mechanism analysis, common benchmark, controlled overlap experiment, and reproducibility safeguards.

3.1 Datasets Definition and Description

3.1.1 Vigitel health survey

Vigitel is Brazil’s telephone-based surveillance system for risk and protective factors for chronic non-communicable diseases [23]. The obesity experiments use a reproducible stratified sample of 5,000 rows drawn from a 2006–2023 extract, comprising 4,149 non-obese and 851 obese records (imbalance ratio approximately 4.9:1). The factorial experiment uses 15 predictors: age, sex, current and former smoking, physical inactivity, leisure-time physical activity, diabetes, hypertension, dyslipidemia, dietary indicators, fruit and vegetable regularity, and schooling. Not every variable in the source extract is used; identifiers, the outcome, and variables excluded by the prespecified analytic pipeline are omitted.

3.1.2 COVID-19 hospital mortality

The COVID-19 dataset contains 334 hospital records and 34 columns, including the binary target `obito` (death). The executed experiments report 251 survivors and 83 deaths, an imbalance ratio of approximately 3.0:1. Predictors include age, sex, alcohol and tobacco indicators, admission-reason groups, number of comorbidities, and specific conditions such as hypertension, heart failure, coronary disease, chronic respiratory disease, diabetes, chronic kidney disease, cancer, cirrhosis, and HIV.

The small sample size and the clinical heterogeneity of the death class are scientifically important rather than incidental. A synthetic-selection method can appear to improve separability simply by concentrating on an artificially homogeneous subset of deaths; results on this dataset must therefore be interpreted with wide uncertainty and assessed strictly on held-out real folds.

3.1.3 Vigitel current smoking

The third task uses the supplied 21,764-record Vigitel analytic sample with current smoking (`fumante`) as the outcome. The executed experiment uses a reproducible stratified sample of 5,000 records: 616 current smokers and 4,384 nonsmokers. Identifiers, colour labels, and upstream cluster fields are excluded; 27 demographic, socioeconomic, diet, activity, and chronic-condition variables are retained. Because the supplied file has no survey-year or sampling-weight field, results describe the analytic sample rather than a weighted population parameter.

3.1.4 Kidney-cohort 365-day mortality

The fourth task is derived from 9,559,286 dated events for 67,267 patients. Each patient’s landmark is 365 days after first recorded activity, and 15 predictor counts are computed strictly before that date. The outcome is recorded death during the subsequent 365 days. Patients who died before the landmark are excluded; nondeaths require observable activity through the horizon. The eligible cohort has a 5.34% event rate; the executed stratified sample contains 5,000 patients, 267 deaths and 4,733 nondeaths. Dialysis is not an outcome in this study.

Table 2 summarizes the sample sizes, class counts, and experimental roles of all four tasks.

Table 2: Datasets used in the reported experiments. Counts are taken from executed repository outputs.

Dataset	n	Positive	Negative	Primary use
Vigitel obesity sample	5,000	851	4,149	SEPAWARE mechanism and augmentation experiments
COVID-19 mortality	334	83	251	SEPAWARE mechanism and augmentation experiments
Vigitel smoking sample	5,000	616	4,384	Common augmentation and TSTR benchmark
Kidney 365-day mortality	5,000	267	4,733	Landmark augmentation validation

3.2 Preprocessing and Cross-validation Protocol

All predictive experiments use stratified five-fold cross-validation. Binary variables are normalized to $\{0, 1\}$, and 1/2-coded survey fields are mapped deterministically. Within each fold, numerical missing values are median-imputed, binary fields use the modal category, and continuous variables are standardized to zero mean and unit variance using statistics fitted only on that fold’s real training subset. CTGAN fitting, candidate scoring, selection, and classifier training all occur within the training portion; the held-out real fold is used only for evaluation.

Every predictive result reported in Section 4 follows this fold-contained protocol. Within each fold of the factorial experiment, a single CTGAN model generates a candidate pool five times the real training size, and the same pool is scored under all four selection conditions described in Section 3.8. Pairing removes generator-to-generator variation between conditions, but the five

fold scores are not independent replications: training folds overlap by 75%, and the four conditions within a fold share a candidate pool. Section 5 discusses this dependence, and the primary factorial model blocks on fold.

We distinguish exploratory from confirmatory evidence throughout. All numerical claims in Sections 4 and 5 come from executed analyses and exported result tables; protocol-only, superseded single-split, or failed runs are not reported as evidence (Appendix A). Results from different evaluation regimes are not pooled because standalone synthetic substitution, real-plus-synthetic augmentation, and internal synthetic cross-validation answer different questions.

3.3 Evaluation Regimes

Three regimes are kept distinct throughout:

Real→Real (TRTR):

train on four real folds and evaluate on the fifth real fold, this is the predictive reference.

Synthetic→Real (TSTR):

train only on synthetic records generated from a real training fold and test on the corresponding held-out real fold, this measures substitution utility.

Real+Synthetic→Real:

append synthetic positive-class records to the real training fold and test on held-out real data; this measures augmentation, not synthetic substitution.

Macro-F1 is the endpoint of the original mechanism experiments because both outcome classes contribute equally. In the common augmentation analysis, AUPRC is the prespecified primary endpoint. Secondary outcomes are Macro-F1, positive-class F1, precision and recall, AUROC, balanced accuracy, Brier score, and calibration intercept and slope. Metrics not produced by the original executed pipeline are reported as unavailable rather than retrospectively reconstructed.

3.4 Generators and Predictive Classifiers

CTGAN [12] is the generator in the mechanism experiments. The common augmentation analysis adds TVAE [12] to test whether the selection mechanism transfers beyond a GAN. Standard-generator and SEPAWARE conditions share a candidate pool within a fold. Because Synthetic Data Vault (SDV) version 1.17 does not implement native conditional sampling for TVAE, TVAE is fitted directly to positive-class training-fold feature records and the outcome label is assigned by design; this is reported separately from CTGAN’s outcome-conditional sampling. Predictive models in the common analysis are logistic regression, random forest, and CatBoost [24]; the mechanism experiments also use XGBoost [25] and TabPFN [26].

3.5 SepAware Selection Framework

Within one fold, let $\mathcal{D}_R = \{(x_i, y_i)\}_{i=1}^n$ denote the real training data and $\mathcal{C} = \{(\tilde{x}_j, \tilde{y}_j)\}_{j=1}^m$ an oversized synthetic candidate pool. Numerical fields are median-imputed, standardized using the training-fold mean and standard deviation, and concatenated with one-hot encodings of categorical fields. Denote this fitted mixed-type representation by $\phi(x)$. All quantities below are fitted from $\mathcal{D}_R$ and evaluated on $\mathcal{C}$; the held-out fold is not used.

For candidate j , the nearest-real distance is

$$d_j = \min_i \|\phi(\tilde{x}_j) - \phi(x_i)\|_2. \quad (1)$$

For a vector z , robust normalization uses $c = \text{median}(z)$ and $s = 1.4826 \text{median}|z - c|$. If $s \leq 10^{-12}$, $s = \text{IQR}(z)/1.349$; if that value is also at most 10^{-12} , every normalized value is set to 0.5. Otherwise,

$$N(z_j) = \frac{\text{clip}((z_j - c)/s, -3, 3) + 3}{6}. \quad (2)$$

The realism score is $R_j = 1 - N(d_j)$ and lies in $[0, 1]$.

Let h_j be the distance to the nearest real record with label $\tilde{y}_j$, and let q_j be the distance to the nearest real record with the opposite label. The geometric margin is $g_j = q_j - h_j$. A random forest fitted only to $\mathcal{D}_R$ gives assigned-label probability $p_j = \hat{P}(Y = \tilde{y}_j \mid \tilde{x}_j)$. The separability score is

$$\text{Sep}_j = \frac{1}{2}N(g_j) + \frac{1}{2}N(p_j), \quad (3)$$

which also lies in $[0, 1]$. When independently prespecified anchors exist, $A_j \in [0, 1]$ is the mean of their class-compatible indicators; otherwise $A_j = 0$ and $\lambda_{\text{anchor}} = 0$. The composite score is

$$S_j = R_j + \lambda_{\text{sep}}\text{Sep}_j + \lambda_{\text{anchor}}A_j, \quad (4)$$

with $\lambda_{\text{sep}} = 1$ and $\lambda_{\text{anchor}} = 0$ in the common benchmark. Within each class, candidates are ordered by decreasing S_j and then by their original candidate index, which provides deterministic tie-breaking. The first Q_y candidates are retained for class y . If fewer than Q_y candidates exist, execution stops with a shortfall error rather than duplicating records. Exact nearest-neighbour scoring requires $O(mn)$ distance evaluations and $O(m \log m)$ sorting, apart from classifier fitting.

Algorithm 1: SepAware candidate selection within one training fold

Inputs: real training data $\mathcal{D}_R$; fitted generator or candidate pool $\mathcal{C}$; target counts Q_y ; feature-type metadata; weights $\lambda_{\text{sep}}, \lambda_{\text{anchor}}$; random seed.

1. Fit imputation, numerical scaling, and categorical encoding on $\mathcal{D}_R$.
2. If necessary, generate a candidate pool larger than the requested class totals using the recorded seed.
3. Fit the scoring random forest on the represented real training data.
4. For every candidate, compute nearest-real distance and R_j .
5. Compute nearest-hit h_j , nearest-miss q_j , margin g_j , and assigned-label probability p_j .
6. Apply $N(\cdot)$ separately to distance, margin, and probability; form Sep_j .
7. If prespecified anchors apply, calculate and normalize A_j ; otherwise set its weight to zero.
8. Form S_j ; rank within class by $(-S_j, j)$; select the first Q_y records or raise a shortfall error.

Outputs: selected synthetic records; component and composite scores; selected candidate indices; class counts and shortfall diagnostics.

Figure 1: Fold-contained implementation of SEPARE. Every fitted quantity is reset in each training fold.

The earlier fixed- λ grid used $\lambda_{\text{sep}} \in \{0, 0.25, 0.5, 1\}$ and $\lambda_{\text{anchor}} \in \{0, 0.25, 0.5\}$. Because configurations were compared on the reporting folds, that grid is exploratory and does not determine the common-benchmark setting. The dataset-specific factorial experiment used predeclared levels $\lambda_{\text{sep}} \in \{0, 1\}$ and $\lambda_{\text{anchor}} \in \{0, 0.5\}$.

Figure 2 summarizes the pipeline. CTGAN produces candidates but does not itself optimize the three selection criteria; a separate selector scores and ranks candidates within each class and returns the requested class counts. Separating generation from selection makes the intervention auditable and allows component-wise ablation, which is the basis of the factorial design in Section 3.8. The design rules out one simple alternative explanation – that a TSTR change is caused by a different class ratio – because class counts are held fixed across all compared conditions. It does not rule out harder alternatives: ranking can select unusually easy prototypes, discard boundary cases, or amplify a classifier-specific shortcut. These possibilities motivate the confirmatory analyses proposed in Section 6.

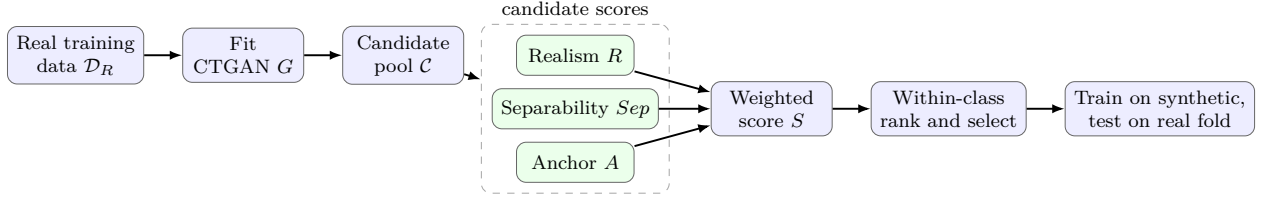


Figure 2: SEPAWARE post-generation selection pipeline: every operation is fitted or computed using the real training portion of a cross-validation fold; the held-out real fold is used only for evaluation.

3.6 Fidelity, Separability, and Privacy Proxies

Fidelity combines numerical-distribution similarity, categorical-proportion similarity, and correlation preservation into a single composite score. Separability is reported using cosine silhouette, a one-nearest-neighbor label-agreement index (SI), and hypothesis margin (HM). These are descriptive geometric measures: higher separation is not automatically clinically desirable, and we do not treat it as such (Section 5).

The common benchmark measures held-out-real positive-class coverage as the mean distance from each positive test record to its nearest synthetic positive record after training-fold standardization. Positive-class diversity is the mean pairwise standardized distance among up to 1000 selected positive records. Exact overlap is the proportion of synthetic records with zero distance to a real training record. The membership-inference proxy compares each real training and held-out record’s distance to the nearest synthetic record and reports membership-classification AUROC. These are empirical diagnostics, not evidence of anonymity or a formal guarantee.

3.7 Domain-anchored Causal Plausibility Filtering

DACPF (Domain-Anchored Causal Plausibility Filtering) generates positive-class CTGAN candidates and ranks their consistency with real positive-class distributions for domain anchors. The completed COVID-19 experiment compares no augmentation, standard CTGAN augmentation, age filtering, disease-anchor filtering, and combined filtering, evaluated as real-plus-synthetic augmentation (Real+Synthetic→Real) rather than standalone synthetic substitution. We emphasize that DACPF is a plausibility filter, not causal discovery or effect estimation: “causal” describes the clinical origin of the anchor hypotheses, not a claim that the filtering step identifies a causal effect.

3.8 Blocked 2^2 Factorial Experiment

The mechanism study uses a two-factor, two-level factorial design: separability weighting is factor A , clinical-anchor weighting is factor B , and all four combinations are evaluated on the same fold-

specific candidate pool. Fold is included as a block because conditions within a fold are paired rather than independent replications.

Let $A \in \{-1, +1\}$ encode separability weighting and $B \in \{-1, +1\}$ anchor weighting. The fold-level response is the classifier-averaged TSTR Macro-F1,

$$y_{ijk} = \mu + q_A A_i + q_B B_j + q_{AB} A_i B_j + \gamma_k + \varepsilon_{ijk}, \quad (5)$$

where y_{ijk} is the classifier-averaged Macro-F1 for separability level i , anchor level j , and fold k ; μ is the grand mean; A_i and B_j are the coded factor levels; q_A , q_B , and q_{AB} are the separability, anchor, and interaction coefficients; γ_k is the fixed effect of fold k ; and ε_{ijk} is residual error. The high-minus-low factorial effects are $2q_A$, $2q_B$, and $2q_{AB}$. Type-II ANOVA partitions sums of squares among factors, interaction, fold, and residual error; the primary variation percentages exclude the fold block from the denominator so that factor contributions are compared against experimental error rather than against total variation. An unblocked $2^2 r$ decomposition that treats folds as replications is also reported in the supplementary material for comparison, but Equation 5 is the more defensible model for paired cross-validation conditions and is used for all primary conclusions.

Both raw and log-transformed responses are analysed. Residual normality is assessed using Shapiro–Wilk and Jarque–Bera tests [27, 28]; constant variance is assessed with median-centred Levene (Brown–Forsythe), Fligner–Killeen, and Breusch–Pagan tests [29–31]. Durbin–Watson and lag-one Ljung–Box statistics diagnose ordering-related serial patterns but cannot establish independence [32, 33]. As noted in Section 3.2, five-fold training sets overlap by 75% and the four conditions within a fold are paired, so inference from this design is treated as exploratory rather than as a formal, independently replicated hypothesis test (Section 5).

3.9 Common Four-task Augmentation and TSTR Experiments

The augmentation experiment uses three repeated five-fold stratified partitions and compares real-only training, random oversampling, random undersampling, SMOTE, class-weighted training, standard CTGAN, SEPAWARE CTGAN, standard TVAE, and SEPAWARE TVAE. Synthetic conditions add positive candidates until the real training fold is balanced, the held-out fold retains its natural prevalence. All missing-value imputation is fitted on the real training fold before resampling or generation. The SEPAWARE augmentation score uses realism plus separability with a fixed unit separability weight, no held-out outcome is used to tune the setting.

Logistic regression supplies a linear and calibration-sensitive reference, random forest supplies a nonlinear ensemble, and CatBoost supplies a strong tabular learner. Hyperparameters and the 0.5 classification threshold are held fixed across methods. Fold-level results are retained so generator comparisons can be paired within dataset, repeated partition, fold, and classifier. The manuscript uses three independently shuffled five-fold partitions with independent fold-specific generator seeds. These repetitions improve stability assessment but remain internal repeats of the same source samples rather than external validation.

The TSTR experiment uses the same repeated partitions. For each generator, standard and selected full synthetic training sets reproduce the real training-fold class counts and share one candidate pool. TSTR results are never pooled with augmentation results.

3.10 Controlled Prevalence-by-overlap Experiment

A separate mechanism experiment varies class prevalence and overlap independently and is not counted as a third health dataset. We generated 2000-record, 15-feature binary-classification problems with 8 informative features, 3 redundant features, 2 clusters per class, and 2% label noise.

Prevalence was set to 5%, 15%, or 30%, and the class separation to 0.4, 1.0, or 2.0, corresponding respectively to high, moderate, and low overlap. Each of the 9 cells used 10 independent data seeds and a stratified 70/30 train-test split.

The comparison includes real-only training, random oversampling, SMOTE, and a SepAware-SMOTE condition. For the latter, SMOTE generates five times the number of required positive-class candidates, the selector ranks them by nearest-real realism, nearest-hit/nearest-miss margin, and random-forest label confidence, then selects enough records to balance the training data. Logistic regression and random forest are evaluated with AUPRC, Macro-F1, and positive-class F1. A heteroscedasticity-robust linear model tests method-by-overlap and method-by-prevalence interactions.

3.11 Ethical considerations

The Vigitel analysis uses a secondary survey extract and the COVID-19 analysis uses a de-identified institutional dataset supplied for the qualification research. The manuscript does not infer a new ethics approval number or consent determination. The authors must insert the applicable institutional review board or ethics committee name, protocol identifier, consent or waiver determination, and any access restrictions before submission.

3.12 Reproducibility

The common experiments were executed with Python 3.12, NumPy 1.26.4, pandas 2.2.2, scikit-learn 1.6.1, SDV 1.17.2, CatBoost 1.2.10, XGBoost 3.2.0, imbalanced-learn 0.12.3, and statsmodels 0.14.2. A project-local environment pins NumPy 1.26.4 because the available compiled scientific extensions were incompatible with NumPy 2. Full software-stack notes, executed-notebook identifiers, and artefact-status decisions are given in the code and data availability statement and in Appendix A.

4 Results

4.1 Dataset and protocol overview

The common benchmark included four outcome tasks from three sources. Vigitel obesity and current smoking share the Vigitel source and are not independent cohorts. The analytic samples contained 851 of 5,000 positive obesity records, 83 deaths among 334 COVID-19 admissions, 616 of 5,000 current smokers, and 267 deaths among 5,000 kidney-cohort patients. Every method used the same held-out real folds within a task. The augmentation export contains 1,620 fold-classifier rows (4 tasks, 9 methods, 3 classifiers, 3 repetitions, and 5 folds); no result was aggregated before export.

4.2 Real-only and conventional imbalance baselines

Table 3 reports the full augmentation comparison. Real-only training had the highest mean AUPRC for Vigitel obesity (0.287), Vigitel smoking (0.300), and kidney mortality (0.122). Random oversampling had the highest COVID-19 AUPRC (0.355). Threshold-dependent conclusions differed: for kidney mortality, real-only training had mean positive-class recall below 0.001 at the fixed 0.5 threshold, whereas random undersampling increased recall to 0.617 but increased the Brier score from 0.050 to 0.225. These results show why ranking, classification at a fixed threshold, and calibration are reported separately.

Table 3: Common augmentation benchmark. Values are means over 3 repeated 5-fold partitions and 3 classifiers (45 paired fold-classifier observations per method and task). Held-out real folds retain natural prevalence; uncertainty and paired inference are reported separately.

Task	Method	AUPRC	Macro-F1	Recall	Brier	n
COVID-19	Class weighting	0.348	0.530	0.373	0.234	45
COVID-19	Random oversampling	0.355	0.538	0.374	0.231	45
COVID-19	Random undersampling	0.332	0.500	0.512	0.279	45
COVID-19	Real only	0.351	0.493	0.113	0.198	45
COVID-19	SMOTE	0.326	0.507	0.204	0.218	45
COVID-19	SepAware CTGAN	0.323	0.534	0.362	0.239	45
COVID-19	SepAware TVAE	0.343	0.508	0.178	0.215	45
COVID-19	Standard CTGAN	0.311	0.521	0.355	0.240	45
COVID-19	Standard TVAE	0.350	0.510	0.183	0.214	45
Kidney mortality	Class weighting	0.106	0.523	0.371	0.156	45
Kidney mortality	Random oversampling	0.103	0.528	0.367	0.155	45
Kidney mortality	Random undersampling	0.117	0.487	0.617	0.225	45
Kidney mortality	Real only	0.122	0.487	0.000	0.050	45
Kidney mortality	SMOTE	0.096	0.520	0.256	0.131	45
Kidney mortality	SepAware CTGAN	0.110	0.526	0.175	0.099	45
Kidney mortality	SepAware TVAE	0.103	0.508	0.054	0.065	45
Kidney mortality	Standard CTGAN	0.104	0.522	0.140	0.094	45
Kidney mortality	Standard TVAE	0.103	0.503	0.042	0.062	45
Vigitel	Class weighting	0.284	0.572	0.476	0.201	45
Vigitel	Random oversampling	0.281	0.573	0.486	0.202	45
Vigitel	Random undersampling	0.277	0.551	0.618	0.230	45
Vigitel	Real only	0.287	0.462	0.009	0.134	45
Vigitel	SMOTE	0.233	0.503	0.204	0.176	45
Vigitel	SepAware CTGAN	0.256	0.527	0.254	0.175	45
Vigitel	SepAware TVAE	0.252	0.492	0.089	0.153	45
Vigitel	Standard CTGAN	0.247	0.511	0.219	0.176	45
Vigitel	Standard TVAE	0.253	0.494	0.096	0.153	45
Vigitel smoking	Class weighting	0.295	0.574	0.452	0.165	45
Vigitel smoking	Random oversampling	0.291	0.582	0.467	0.165	45

Task	Method	AUPRC	Macro-F1	Recall	Brier	n
Vigitel smoking	Random undersampling	0.284	0.559	0.654	0.210	45
Vigitel smoking	Real only	0.300	0.500	0.036	0.099	45
Vigitel smoking	SMOTE	0.229	0.541	0.192	0.134	45
Vigitel smoking	SepAware CTGAN	0.249	0.565	0.288	0.143	45
Vigitel smoking	SepAware TVAE	0.255	0.526	0.099	0.111	45
Vigitel smoking	Standard CTGAN	0.217	0.536	0.216	0.145	45
Vigitel smoking	Standard TVAE	0.253	0.523	0.095	0.111	45

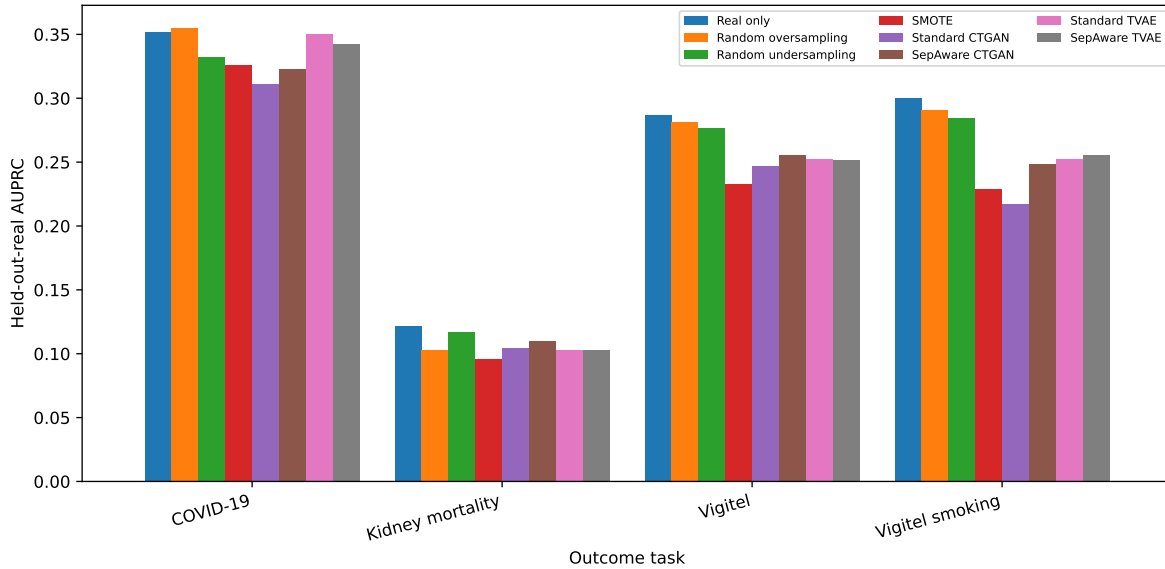


Figure 3: Held-out-real AUPRC in the augmentation regime. Bars are means over 3 repeated 5-fold partitions and 3 classifiers. Methods within each task use identical test folds; corrected paired comparisons are shown in Figure 4.

4.3 Standard generators and SepAware selection

For CTGAN, mean AUPRC differences between SEPAWARE and standard sampling were 0.012 for COVID-19, 0.009 for Vigitel obesity, 0.031 for Vigitel smoking, and 0.005 for kidney mortality. The Nadeau–Bengio corrected intervals were $[-0.055, 0.079]$, $[-0.004, 0.021]$, $[0.015, 0.048]$, and $[-0.007, 0.017]$, respectively. Only smoking remained significant after Holm adjustment ($p_{\text{adj}} = .009$). TVAE differences ranged from -0.007 to 0.003 , and all corrected intervals included zero. The inferential unit was the mean difference across classifiers within each repeated-CV fold ($n = 15$ fold summaries per contrast), not 45 fold-classifier observations.

The selected CTGAN condition did not surpass the strongest conventional method on any task. Its mean AUPRC was 0.323 for COVID-19, 0.256 for Vigitel obesity, 0.249 for Vigitel smoking, and 0.110 for kidney mortality. The corresponding best conventional values were 0.355, 0.287, 0.300, and 0.122. SepAware therefore improved one generator comparison reliably without establishing superiority over simpler training strategies.

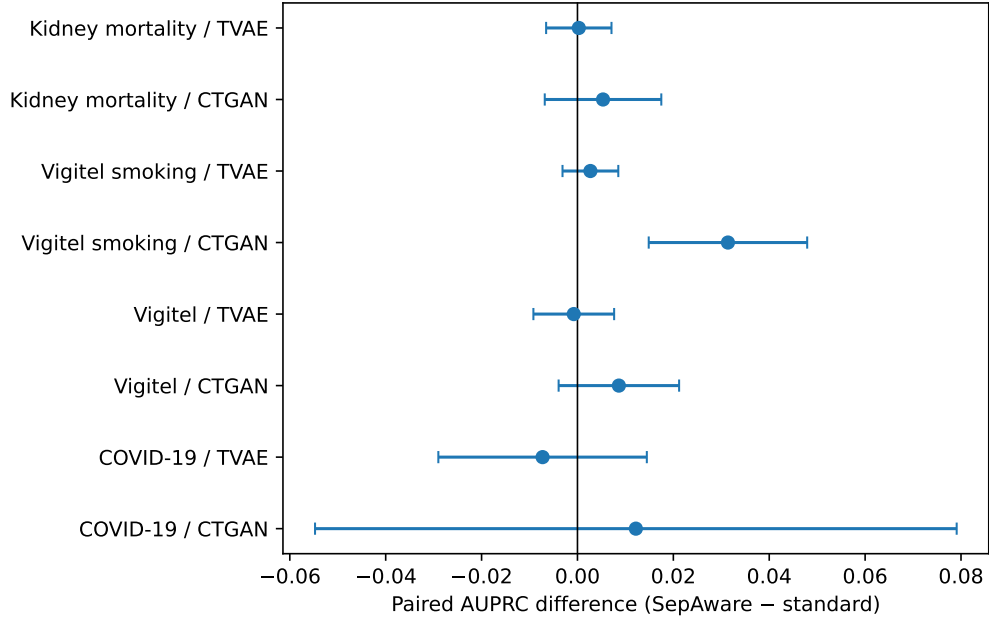


Figure 4: Primary paired AUPRC contrasts in real-plus-synthetic augmentation. Points are mean SEPAWARE minus standard-generator differences; intervals use the Nadeau–Bengio repeated-cross-validation correction. Eight task-generator tests were adjusted together by Holm’s method.

4.4 TSTR utility

The common TSTR experiment trained each classifier only on a synthetic dataset whose class counts matched the corresponding real training fold and evaluated on the untouched real test fold. Standard and selected variants shared candidate pools. Table 4 reports all tasks and both generators; these substitution results are kept separate from augmentation.

Table 4: TSTR performance. Values are means over 3 repeated 5-fold partitions and 3 classifiers (45 paired observations per condition and task). Synthetic class counts match each real training fold; evaluation uses held-out real records.

Task	Synthetic training data	AUPRC	Macro-F1	Recall	Brier
COVID-19 hospital mortality	SepAware CTGAN	0.333	0.517	0.194	0.229
COVID-19 hospital mortality	SepAware TVAE	0.324	0.478	0.498	0.385
COVID-19 hospital mortality	Standard CTGAN	0.279	0.459	0.068	0.214
COVID-19 hospital mortality	Standard TVAE	0.332	0.491	0.468	0.347
Kidney-cohort 365-day mortality	SepAware CTGAN	0.080	0.491	0.007	0.059

Task		Synthetic training data	AUPRC	Macro-F1	Recall	Brier
Kidney-cohort 365-day mortality		SepAware TVAE	0.100	0.519	0.368	0.176
Kidney-cohort 365-day mortality		Standard CTGAN	0.063	0.486	0.000	0.052
Kidney-cohort 365-day mortality		Standard TVAE	0.102	0.524	0.336	0.160
Vigitel smoking	current	SepAware CTGAN	0.279	0.601	0.319	0.128
Vigitel smoking	current	SepAware TVAE	0.179	0.443	0.689	0.419
Vigitel smoking	current	Standard CTGAN	0.139	0.467	0.000	0.109
Vigitel smoking	current	Standard TVAE	0.183	0.428	0.700	0.432
Vigitel obesity		SepAware CTGAN	0.279	0.548	0.175	0.148
Vigitel obesity		SepAware TVAE	0.200	0.348	0.847	0.572
Vigitel obesity		Standard CTGAN	0.191	0.454	0.001	0.143
Vigitel obesity		Standard TVAE	0.223	0.502	0.447	0.308

For CTGAN, selection increased mean TSTR AUPRC from 0.279 to 0.333 for COVID-19, 0.191 to 0.279 for Vigitel obesity, 0.139 to 0.279 for smoking, and 0.063 to 0.080 for kidney mortality. TVAE changes were smaller and mixed: 0.332 to 0.324, 0.223 to 0.200, 0.183 to 0.179, and 0.102 to 0.100, respectively. These are descriptive substitution results and do not replace the prespecified augmentation contrasts.

4.5 Calibration, coverage, diversity, fidelity, and privacy diagnostics

In augmentation, CTGAN selection reduced positive-class coverage distance for every task: from 4.410 to 4.115 for COVID-19, 1.977 to 1.723 for Vigitel obesity, 3.833 to 3.586 for smoking, and 1.744 to 1.595 for kidney mortality. It also reduced within-positive-class diversity from 11.332 to 7.888, 5.478 to 4.356, 7.340 to 6.310, and 6.827 to 4.425, respectively. Thus closer coverage was consistently accompanied by a more concentrated selected sample.

In augmentation, exact-match rates were zero or below 0.0005, and membership-proxy AUROCs ranged from 0.499 to 0.508 for standard and selected CTGAN. In TSTR, kidney TVAE had much higher exact-match rates: 0.105 for standard TVAE and 0.134 after selection. TSTR detection AUROC was also high for several conditions, reaching 0.995–0.998 for kidney CTGAN and at least 0.979 for kidney TVAE. These results show readily detectable distributional differences and, for kidney TVAE, material duplication risk. Membership-proxy AUROCs remained near 0.5, but none of these measures establishes anonymity. The full marginal similarity, correlation similarity, detection, coverage, diversity, exact-match, and membership-proxy matrix is provided as a machine-readable supplementary table.

4.6 Dataset-specific anchor and factorial analyses

Clinical-anchor analyses were not part of the common benchmark. The COVID-19 DACPF experiment required age, cancer, and respiratory-admission variables; comparable anchors were not

prespecified for smoking or kidney mortality. Applying invented substitutes after seeing the data would have introduced outcome-guided tuning.

In the five-fold COVID-19 DACPF augmentation run, disease-anchor filtering produced mean Macro-F1 of 0.577 for CatBoost and 0.534 for XGBoost, compared with 0.526 for each classifier under standard CTGAN augmentation. Age-only and combined filters did not improve on standard augmentation. The result is dataset- and classifier-specific.

The blocked 2^2 anchor-by-separability TSTR analysis was likewise limited to Vigitel obesity and COVID-19. The fold-paired Vigitel separability effect was 0.102 Macro-F1 (95% CI 0.077–0.128; exploratory $p < .001$). The COVID-19 estimate was -0.018 (95% CI -0.063 – 0.027 ; $p = .332$). Anchor and interaction intervals included zero for both tasks. Overlapping training folds and shared candidate pools limit these exploratory intervals; they are not independent replications.

4.7 Controlled simulation

The simulation varied prevalence and class overlap independently and is not counted as a health-data task. SepAware-SMOTE exceeded ordinary SMOTE in each of nine prevalence–overlap cells for logistic regression, but mean differences were small (0.001–0.024). In pooled heteroscedasticity-robust regression, neither the method contrast nor its overlap and prevalence interactions was distinguishable from zero (all $p > .91$). The simulation therefore supports separating prevalence from overlap but does not locate a stable region in which selection improves predictive performance.

4.8 Sensitivity and statistical interpretation

Classifier-by-repetition effects are exported with the primary statistics. Their variation is consistent with the wide corrected intervals for seven of eight primary contrasts. Results from Vigitel obesity and smoking cannot be treated as two independent source replications. The fixed- λ grid is retained only as an exploratory development record because configurations were compared on the same folds used for reporting.

5 Discussion

5.1 Interpretation of the common benchmark

SepAware selection produced a corrected AUPRC gain over standard CTGAN for Vigitel smoking. The other three CTGAN intervals and all four TVAE intervals included zero after accounting for repeated-cross-validation dependence, and the selected generator did not exceed the strongest conventional method on any task. The result is narrower than a claim that separability-aware generation improves rare-outcome prediction: selection can improve a specific candidate pool, but it did not displace real-only training or resampling as the practical reference.

The distinction between TSTR and augmentation is material. TSTR asks whether a synthetic dataset retains enough information to replace real training data. Augmentation asks whether synthetic positive records add information when the real observations remain available. A method can improve substitution utility relative to its generator while adding little in augmentation. We therefore report the regimes separately and avoid averaging across them.

The two Vigitel outcomes share a source. Agreement between their estimates may reflect common measurement, preprocessing, or population structure and cannot be interpreted as replication in an independent cohort. COVID-19 mortality has a small sample and broad uncertainty; kidney

mortality has lower prevalence and a longitudinally derived feature set. This heterogeneity is useful for observing failure, but four outcome tasks do not establish transportability.

5.2 Selection, coverage, and diversity

CTGAN selection moved synthetic positive records closer to held-out positive observations in every task. At the same time, pairwise diversity decreased in every task. These changes are compatible with two explanations: the selector may retain clearer class-conditional signal, or it may discard legitimate boundary cases. The available labels cannot determine which removed variation is measurement noise, redundant spread, or clinically meaningful heterogeneity.

The kidney task illustrates the limitation. Coverage distance improved and diversity declined, yet its corrected CTGAN AUPRC interval included zero and real-only training retained the highest mean AUPRC. Geometry changed as designed without producing a demonstrated ranking benefit. A useful deployment criterion must therefore include predictive performance against simple baselines, calibration, and clinically interpretable coverage rather than the composite selection score alone.

TVAE also shows dependence on candidate-pool quality. Under SDV 1.17.2, class-specific models were required because native outcome-conditional sampling was unavailable. Their positive-class diversity was low, and subsequent selection did not create information absent from the pool. The scoring rule can be applied to different generators, but its benefit is not generator independent.

5.3 Dataset-specific mechanism evidence

The Vigitel-obesity factorial estimate attributes a Macro-F1 change to separability weighting within that executed design; the corresponding COVID-19 interval includes zero. The factorial results reuse overlapping folds and shared candidate pools, so their small-sample intervals are exploratory. The anchor component is also task-specific. COVID-19 anchors were defined from available clinical variables, whereas defensible analogues were not prespecified for smoking or kidney mortality. Extending those anchors by analogy after observing outcomes would weaken, not strengthen, the comparison.

The controlled simulation separates overlap from prevalence but does not show a significant method-by-overlap interaction. It supports the design premise that imbalance and overlap are different properties; it does not establish a threshold at which SepAware should be used.

5.4 Calibration and operating thresholds

The kidney results show that a model can have useful ranking while predicting no positives at a fixed 0.5 threshold. Undersampling increased recall but substantially worsened the Brier score. Threshold selection should therefore depend on the intended clinical operating point and asymmetric error costs. This study fixed the threshold and did not tune it on test folds; it does not recommend a deployment threshold.

5.5 Privacy interpretation

Augmentation exact-match rates were low, but standalone kidney TVAE data had exact-match rates of 10.5% under standard sampling and 13.4% after selection. Several real-versus-synthetic detection AUROCs were also close to one. These findings argue against releasing those synthetic data. Membership-proxy AUROCs near 0.5 do not offset that concern: the proxy attacker uses

nearest-synthetic distance and does not represent stronger shadow-model, linkage, or attribute-inference attacks. Disclosure risk must be assessed under an explicit attacker model before any synthetic records are released.

5.6 Relation to prior work

The competitiveness of random oversampling agrees with evidence that sampling with replacement can match more complex augmentation in small-data settings [18]. Our results add a within-generator comparison: selecting candidates can change utility and geometry without making synthetic augmentation the best strategy. Evaluation frameworks that separate fidelity, utility, and privacy are therefore more informative than a single global quality score [15, 16].

The study supports treating separability as a measurable intervention rather than an assumed virtue. Its practical value depends on the task, generator, candidate pool, classifier, and acceptable loss of positive-class heterogeneity. External validation and phenotype-aware coverage analyses are needed before the method can support a clinical claim.

6 Limitations

Four outcome tasks from three health-data sources were evaluated. This broadens the observed prevalence and outcome contexts but does not identify a general relationship between intrinsic overlap and SepAware benefit; the two Vigitel tasks share a source and are not independent cohorts.

The Vigitel factorial finding comes from one 5000-record sample, one cross-validation partition, one CTGAN configuration, and shared within-fold candidate pools. The common benchmark uses three repeated five-fold partitions and 30 training epochs. Fold-specific generator seeds avoid reusing a single fitted generator, and repeated shuffles measure partition variability, but all repetitions reuse the same source observations. More generator seeds, higher-capacity sensitivity runs, and independent datasets are necessary before confirmatory claims.

The fixed lambda grid was exploratory because performance and configuration selection used the same folds. The common analysis fixes the separability weight in advance, but it does not estimate a clinically optimal trade-off between discrimination and positive-class diversity. The score also depends on a random-forest confidence term; additional boundary models and label-permutation controls are needed to test classifier-specific shortcuts.

The TVAE condition is not directly symmetric with CTGAN. SDV 1.17 does not implement native TVAE conditional sampling, so TVAE was fitted to positive-class features. This is a reproducible conditional-generation strategy, but its low diversity shows that post-selection cannot compensate for a collapsed or highly concentrated candidate pool. A diffusion model and a generator with native conditional support should be added before making a broad generator-agnostic claim.

Positive-class coverage and diversity are geometric proxies rather than clinically validated phenotype measures. The COVID-19 death class may contain several pathways that are not represented by unsupervised distances. Blinded clinician review and prespecified phenotype or subgroup definitions are required to determine whether removed variation is clinically meaningful. Calibration estimates are also unstable in the small COVID-19 folds.

The privacy analysis is limited to exact matches, nearest-record distances, and a membership proxy. It does not include attribute inference, shadow-model membership attacks, linkage attacks, or a formal differential-privacy guarantee. No synthetic data release should be justified from the present privacy results.

Finally, external or temporal validation is absent. The smoking file contains no survey-year field, the obesity evaluation is not a prospective temporal split, and the COVID-19 cohort is institution-specific. Kidney mortality uses a longitudinal landmark but random patient partitions rather than a calendar-time split. At least one temporal or cross-site evaluation is needed.

These limitations define the next additions: a native conditional third generator, multiple generator seeds per fold, expert-informed positive-class coverage, stronger privacy attacks, formal overlap-to-benefit modeling, and external or temporal validation. Failures should continue to be reported as operational guidance about when separability control should not be used.

7 Conclusions

Class imbalance and class overlap are not interchangeable problems. In the dataset-specific factorial experiment, explicit separability pressure changed Vigitel-obesity utility but not COVID-19 utility. In the common augmentation benchmark, the corrected CTGAN comparison supported an AUPRC gain only for Vigitel smoking; the other CTGAN and all TVAE intervals included zero. Simple rebalancing was often competitive, and closer positive-class coverage was accompanied by reduced diversity.

SepAware is an auditable mechanism for controlling and studying class-conditional geometry, not a universally superior augmentation method. Its use requires evidence that a predictive gain survives simple baselines and does not result from removing difficult but legitimate positive cases. Internal repeated cross-validation cannot establish clinical benefit or transportability.

Declarations

Ethics approval and consent to participate

The repository does not contain the responsible ethics-committee names, approval or exemption identifiers, or consent/waiver determinations for the institutional COVID-19 and kidney cohorts. These factual statements cannot be inferred from de-identified analytic files. They must be supplied and verified by the data custodians before submission. No patient-level source data are included in the repository.

Data availability

The source datasets are not distributed with this manuscript. The code repository records required file names, cryptographic hashes, expected schemas, cohort rules, and analytic counts so authorized users can validate their copies. The Vigitel obesity and smoking tasks share a surveillance source; the supplied smoking extract lacks survey-year and sampling-weight metadata. Access routes and data-use restrictions for the institutional cohorts must be confirmed by the authors before submission.

Code availability

The repository contains the experiment manifest, executable pipelines, tests, fold-level results, statistical scripts, and table and figure builders. A permanent public archive URL and software license remain to be assigned after confirming that no restricted source records or credentials are present.

Funding, competing interests, and author contributions

Funding identifiers, competing-interest declarations, and CRediT contributions were not present in the supplied research materials. Author-confirmed statements are required before submission; no declarations have been inferred.

Use of generative tools

Generative tools assisted with code refactoring and language editing. The author remains responsible for verifying the analyses, citations, interpretation, and final text and should adapt this disclosure to the target journal’s policy.

Abbreviations

AUPRC area under the precision-recall curve

AUROC area under the receiver operating characteristic curve

CTGAN conditional tabular generative adversarial network

HM hypothesis margin

SI separability index

SMOTE synthetic minority oversampling technique

TSTR train on synthetic, test on real

TVAE tabular variational autoencoder

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A Supplementary Methods and Results

A.1 Output hierarchy

Fold-level augmentation results are stored in `results/raw/predictive_metrics_long.csv`; TSTR results are stored in `results/raw/tstr_fold_metrics.csv`. Diagnostic files use the same task, repetition, fold, and condition keys. Aggregate tables are generated from these files and are not edited manually. The experiment manifest records source hashes, features, exclusions, cohort construction, seeds, generator settings, metrics, and task-specific applicability.

The common benchmark contains real-only training, random over- and undersampling, SMOTE, class weighting, standard and selected CTGAN, and standard and selected TVAE in augmentation. TSTR contains the four generator conditions. The full metric matrix includes AUPRC, AUROC, Macro-F1, positive-class F1, precision, recall, balanced accuracy, Brier score, calibration intercept, and calibration slope. Machine-readable summaries are in `results/tables/augmentation_metric_summary.csv` and `results/tables/tstr_metric_summary.csv`.

A.2 Repeated-cross-validation inference

For each task and generator, SepAware and standard observations are paired by repetition, fold, and classifier. Classifier differences are averaged within a fold before inference. If $\bar{d}$ and s_d^2 are the mean and variance of the 15 fold summaries, the corrected standard error is

$$\text{SE}_{\text{NB}} = \sqrt{\left(\frac{1}{15} + \frac{n_{\text{test}}}{n_{\text{train}}}\right) s_d^2}, \quad (6)$$

with $n_{\text{test}}/n_{\text{train}} = 0.25$ for five-fold cross-validation. Two-sided t intervals use 14 degrees of freedom. Holm adjustment is applied across the eight prespecified task-generator contrasts. This correction addresses overlap approximately; it does not make internal folds equivalent to independent cohorts.

A.3 Fidelity and disclosure-risk diagnostics

Numerical marginal similarity is one minus the Kolmogorov–Smirnov distance; categorical similarity is one minus total-variation distance. Marginal similarity averages these feature-level values. Correlation similarity is one minus half the mean absolute difference between real and synthetic correlation matrices over the imputed numeric and numeric-coded categorical fields. Detection

AUROC comes from a classifier distinguishing real training records from synthetic records; 0.5 indicates chance discrimination and values farther from 0.5 indicate easier detection.

Coverage is the mean distance from each held-out positive record to its nearest synthetic positive record. Diversity is the mean pairwise distance among at most 1,000 synthetic positive records. Exact-match rate is the synthetic fraction at zero encoded distance from a real training record. The membership proxy classifies real training versus held-out records using nearest-synthetic distance. None of these metrics provides a formal privacy guarantee.

A.4 Dataset-specific analyses

The COVID-19 DACPF files are stored in `results/dataset_specific/`. The factorial fold responses, effects, assumption diagnostics, and correlation analyses are stored in `results/factorial/`. These analyses are not averaged with common-benchmark results. No clinical-anchor experiment was created for smoking or kidney mortality because independently justified anchor definitions were unavailable.

A.5 Excluded and superseded evidence

Single-split development notebooks, failed TabPFN runs, and protocols without successful outputs are excluded from numerical claims. The reporting-fold fixed- λ search is retained as exploratory development evidence but is superseded for the common benchmark by the fixed manifest setting. The repository audit and task-by-experiment matrix give the status and rationale for every identified experiment.